



FRACA SERVCOM INVENTORY MANAGEMENT SYSTEM

SYSTEM REQUIREMENT SPECIFICATION

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1.0 INTRODUCTION

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to describe the functional and non-functional requirements for the FRACA SERVCOM Inventory Management System. This SRS is intended for stakeholders including the client (FRACA SERVCOM management), system developers, system designers, testers, and future maintainers of the application. The document clearly outlines what the system should do and serves as the foundation for development and validation.

1.2 Scope

FRACA SERVCOM Inventory Management System is a web-based system designed to automate stock tracking and record-keeping for furniture and hardware products. The system allows users to add products, update stock, monitor stock levels, manage sales & purchases, and generate basic inventory reports.

The system will:

- Track products and categories
- Record stock quantity and stock changes
- Send low-stock alerts
- Manage purchases and sales
- Generate inventory and activity reports
- Support role-based access (Admin, Staff)

The system will not:

- Support multiple branches
- Provide a mobile native app
- Handle advanced analytics or forecasting
- Integrate with external ERP or POS systems

System objectives:

- Improve accuracy and speed of inventory updates
- Reduce stock-outs through alerts
- Provide simple reporting for decision-making
- Scrap manual paperwork and reduce human error

1.3 Definitions, Acronyms & Abbreviations

- FRACA SERVCOM – The client company using the system
- IMS – Inventory Management System
- SKU – Stock Keeping Unit
- UI – User Interface
- CRUD – Create, Read, Update, Delete
- PHP – Hypertext Preprocessor
- MySQL – Database management system
- HTML/CSS/JS – Web development languages

1.4 Overview

This SRS contains:

Chapter 1: Introduction, purpose, scope, definitions, and SRS overview

Chapter 2: General system description including system perspective, functionalities, users, constraints, assumptions

Chapter 3: Detailed functional and non-functional requirements including inputs, outputs, processes, and interface requirements

Chapter 4: Appendices containing sample data, diagrams, and supporting documents

2.0 **GENERAL DESCRIPTION**

2.1 System Perspective

The Inventory Management System is a stand-alone web-based system operating on a single location within FRACA SERVCOM.

It will consist of:

Major components:

- Frontend (HTML, CSS, JavaScript)
- Backend (PHP / Laravel optional)
- Database (MySQL)

External Interactions:

- Users interact through a web browser
- Database stores all inventory records
- Email or on-screen notifications for alerts

Hardware requirements:

- A standard PC or laptop
- Local server (XAMPP) or hosted web server
- Block Diagram (text-described):

User → Web Browser → PHP Backend → MySQL Database → System Output

2.2 System Functionality

The system will perform the following core functions:

Product Management

- Add, view, edit, and delete product details
- Assign barcode, category, price, and stock thresholds

Stock Management

- Increase/decrease stock levels
- Automatic logging of stock changes
- Low-stock detection and alerts

Sales & Purchase Recording

- Record sales transactions
- Record purchase/stock-in transactions
- Update stock automatically

Reporting

- Generate daily/weekly/monthly stock reports

- View low-stock report
- Export basic data (optional)

User Management

- Admin can create staff accounts
- Role-based privileges

2.3 User Characteristics

Admin

- Moderate computer skills
- Responsible for managing users, products, and system settings
- Requires full access

Inventory Clerk / Staff

- Basic computer literacy
- Daily use of stock updates and transaction recording
- Needs an interface that is simple and intuitive

Management

- Low technical involvement
- Uses the reporting module for decision-making

Each user group has different expectations, mainly around simplicity, accuracy, and speed.

2.4 General Constraints

- System must run on PHP and MySQL
- Users should have stable internet or local network access
- Access control must prevent unauthorized access
- Data modification must follow security policies
- Limited to one branch; no distributed architecture

2.5 Assumptions & Dependencies

- Users have basic knowledge of computer operations
- FRACA SERVCOM provides working PCs and stable electricity
- Accurate product data will be provided during system setup
- The system depends on XAMPP or a similar server environment
- All staff will be trained on how to use the system

3.0 SPECIFIC REQUIREMENTS

3.1 Functional Requirements

Inputs

- Product details (name, barcode, price, category)
- Stock updates (quantity in/out)
- Sales and purchase records
- User login credentials

Input Validation:

- Numeric fields must accept numbers only
- Mandatory fields must be filled before submission

Outputs:

- Updated stock quantities
- Low-stock notifications
- Reports (stock report, sales summary, purchase summary)
- Confirmation messages after transactions

Timing Constraints:

- Reports should generate in under 5 seconds
- Low-stock alerts should appear immediately after stock check

Processes

- Automatic stock deduction after sale
- Validation of all inputs
- Alert generation when threshold is reached
- User authentication and session management
- Error messages on invalid operations

3.2 User Interface Requirements

User Interface

- Login page
- Dashboard with statistics
- Product management pages
- Stock update pages
- Sales/Purchase forms
- Reports page

UI Characteristics:

- Clean and simple layout
- Accessible on desktop screens
- Uses buttons, tables, and forms

Hardware Interface

- Minimum: 4GB RAM, 2.0 GHz processor
- 500 MB free disk space
- Keyboard and mouse for input

Software Interface

- Runs on PHP 8+
- MySQL 5.7 or above
- Works on Windows or Linux servers
- Tested on Chrome and Firefox browsers

Communication Interface

- LAN or local WiFi for internal use
- Optional SMTP for email alerts

Other Requirements

- Automatic backups (if configured by admin)
- System should run reliably during working hours
- Data retention: All transactions stored indefinitely unless archived

4.0 APPENDIX

4.1 Use cases

Use Case 1: **User Login**

Actor: Admin / Staff / Manager

Precondition: User account exists

Trigger: User enters credentials and clicks “Login”

Main Flow:

1. User enters username and password
2. System validates fields
3. System checks credentials against database
4. System creates a session
5. System redirects user to the Dashboard

Postcondition: User is authenticated

Exceptions:

- Incorrect username/password → system displays error
- Account disabled → access denied

Use Case 2: **Add Product**

Actor: Admin / Inventory Staff

Precondition: User is logged in

Main Flow:

1. User selects “Add Product”
2. User enters product details (name, barcode, price, category, stock threshold)
3. System validates fields
4. System creates a new product record in DB
5. System confirms successful creation

Postcondition: Product stored in database

Exceptions:

- Duplicate barcode → reject
- Missing mandatory fields → prompt user

Use Case 3: **Update Stock (Stock-In)**

Actor: Admin / Staff

Main Flow:

1. User selects “Add Stock”

2. User enters product ID and quantity
3. System validates product exists
4. System increases stock balance
5. System logs stock-in transaction
6. System displays confirmation

Exceptions:

- Product not found
- Invalid quantity (negative, zero, or non-numeric)

Use Case 4: Reduce Stock (Stock-Out)

Actor: Admin / Staff

Main Flow:

1. User selects “Reduce Stock”
2. User enters product ID and quantity
3. System validates stock availability
4. System decreases stock balance
5. System logs stock-out transaction
6. System checks threshold
7. If below, system generates low-stock alert

Exceptions:

- Insufficient stock
- Invalid or missing inputs

Use Case 5: **Record Sale**

Actor: Staff

Main Flow:

1. User opens “Record Sale”
2. Selects product and quantity sold

3. System checks stock
4. System deducts stock
5. System creates sale record
6. System auto-updates stock logs
7. System displays receipt-like summary

Exceptions:

- Product not found
- Not enough stock

Use Case 6: **Record Purchase**

Actor: Admin / Staff

Main Flow:

1. User selects “Record Purchase”
2. Enters supplier and product details
3. System validates inputs
4. System updates stock
5. System logs purchase

Use Case 7: Generate Reports

Actor: Admin / Manager

Main Flow:

1. User selects report type (stock, sales, purchases, low-stock)
2. System retrieves relevant data
3. System generates formatted report
4. User views/export report

Exceptions:

- No data available (system shows empty report)

Use Case 8: Manage Users

Actor: Admin

Main Flow:

- Admin selects “User Management”
- Admin creates/edits/deletes a user
- System updates the user table

Exceptions:

- Duplicate username
- Invalid roles

4.2 Survey results

The following data was collected during interviews and observations at FRACA SERVCOM:

Summary of Findings

- Stock updates are done manually using paper
- Staff experience frequent stock-out incidents
- Reports are prepared once per week due to manual delays
- No automated alert system exists